

Ken Boston

RE-2022-520703



United International University

Document Details

Submission ID

trn:oid:::3618:89215360

Submission Date

Apr 2, 2025, 3:20 PM GMT+5:30

Download Date

Apr 2, 2025, 3:24 PM GMT+5:30

File Name

RE-2022-520703.docx

File Size

73.6 KB

6 Pages**2,468 Words****15,737 Characters**

18% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

-  **37 Not Cited or Quoted 15%**
Matches with neither in-text citation nor quotation marks
-  **9 Missing Quotations 3%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 9%  Internet sources
- 9%  Publications
- 13%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 37 Not Cited or Quoted 15%**
Matches with neither in-text citation nor quotation marks
- 9 Missing Quotations 3%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 9% Internet sources
- 9% Publications
- 13% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Harrisburg University of Science and Technology on 2024-10-20	2%
2	Publication	Ravinder Kumar, Lokesh Kumar Shrivastav. "An Ensemble of Random Forest Grad...	2%
3	Submitted works	University of Wales, Bangor on 2024-05-30	1%
4	Internet	www.ijecs.in	1%
5	Submitted works	The University of the West of Scotland on 2023-04-21	<1%
6	Internet	www.ijritcc.org	<1%
7	Internet	www.mdpi.com	<1%
8	Submitted works	Sheffield Hallam University on 2024-09-10	<1%
9	Internet	www.ijariit.com	<1%
10	Submitted works	Chatrabhuj Narsee School on 2024-12-17	<1%

11	Submitted works	University of Greenwich on 2024-04-25	<1%
12	Publication	Liang Zhang, Zhong Cheng, Zhouzheng Hao, Kai Zuo, Kefei Liu. "Integrating inflo...	<1%
13	Submitted works	Athlone Institute of Technology on 2024-08-25	<1%
14	Publication	V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng...	<1%
15	Submitted works	University of West London on 2023-06-04	<1%
16	Internet	finmodelslab.com	<1%
17	Internet	fastercapital.com	<1%
18	Internet	ijip.in	<1%
19	Internet	www.jneonataisurg.com	<1%
20	Internet	ijsra.net	<1%
21	Internet	ijsret.com	<1%
22	Internet	gsarpublishers.com	<1%
23	Internet	rdr.io	<1%
24	Publication	Bui Thanh Hung, M. Sekar, Ayhan Esi, R. Senthil Kumar. "Applications of Mathema...	<1%

25

Submitted works

Southern New Hampshire University - Continuing Education on 2024-10-27

<1%

STOCK PRICE PREDICITON USING MACHINE LEARNING

DR.V.SATHIYA

Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India

S.JANESH KUMAR

Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
janeshkumar973@gmail.com

W.Ken Boston

Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
kenboston398@gmail.com

P.Malai raja

Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
malairajap11@gmail.com

Abstract - Predicting stock prices is difficult because financial markets are complicated and volatile. Prediction accuracy has been increased using a variety of techniques, such as machine learning, deep learning, and statistical methods. Based on past data and market indicators, this study investigates various methods for predicting stock values, including time series analysis, regression models, and neural networks, especially Long Short-Term Memory (LSTM). In addition, elements like trading volume, economic trends, and sentiment research from social media and news sources are taken into account. In terms of identifying market trends and patterns, the results indicate that sophisticated machine learning models perform better than conventional techniques, giving investors important information for making decisions. A number of variables, including past price trends, market sentiment, macroeconomic indicators, and corporate performance, affect the dynamic and difficult problem of stock price prediction. Traditional methods like statistical time series models (ARIMA, GARCH) have been widely used for forecasting but often fail to capture the non-linear and highly volatile nature of stock prices.

I. INTRODUCTION

Stock market prediction has been a widely researched topic due to its financial significance and impact on global economies. Investors, traders, and financial analysts constantly seek methods to forecast stock price movements to maximize returns and minimize risks. Traditional forecasting techniques, such as fundamental and technical analysis, rely on historical price trends, company performance, and economic indicators. However, these methods often struggle to capture the highly volatile and nonlinear nature of financial markets. As artificial intelligence and data driven technologies have advanced, machine learning

(ML) has become a potent tool for predicting stock prices. Massive amounts of historical stock data may be processed by ML algorithms, which can then uncover hidden predicting stock prices. Massive amounts of historical stock data may be processed by ML algorithms, which can then uncover hidden patterns and use learned relationships to forecast future. In financial forecasting, trees, random forests, and deep learning models such as Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks have all been used with success. Moreover, integrating alternative data sources, such as news sentiment analysis, social media trends, and macroeconomic indicators, has further improved prediction accuracy. While challenges such as data noise, market unpredictability, and overfitting remain, machine learning continues to evolve, offering more sophisticated models for stock price forecasting. This study explores various ML techniques applied to stock market prediction, comparing their effectiveness and discussing their potential to enhance investment strategies.

A. Problem Definition

The extremely dynamic, volatile, and non-linear character of financial markets makes stock price prediction a difficult and complex undertaking. Technical analysis and statistical models are two examples of traditional forecasting methods that frequently fall short of accurately predicting market movements because they are unable to identify intricate patterns and correlations in stock price data. Investors and traders require more reliable and data-driven approaches to make informed decisions and minimize financial risks. The primary challenge in stock price prediction is the influence of multiple factors, including historical price trends, trading volume, macroeconomic indicators, news sentiment, and investor behavior. Additionally, external events

. such as economic policies, geopolitical tensions, and global crises further complicate market dynamics. It is challenging to create a model that reliably produces high-accuracy forecasts due to the unpredictable nature of market swings. There is an increasing chance to use computer models that can process big datasets, find hidden patterns, and increase prediction accuracy thanks to the development of machine learning (ML) and deep learning approaches. However, several issues persist, such as data quality, feature selection, overfitting, and model interpretability. Therefore, this study aims to develop prediction, evaluating their effectiveness and limitations to provide investors with more reliable forecasting tools.

II LITERATURE SURVEY

1. Autoregressive Integrated Moving Average (ARIMA):

Because of its ability to capture trends and seasonality, Autoregressive Integrated Moving Average (ARIMA) models have been utilized extensively for stock price forecasting (Box & Jenkins, 1976). Nevertheless, they have trouble with really erratic data. The AR component of ARIMA indicates that the time series has been regressed using its own historical data. According to the MA component of ARIMA, the prediction error is a linear combination of the corresponding historical mistakes. The I component of ARIMA indicates that, in order to achieve stable data—a prerequisite for the ARIMA model approach—the data values have been substituted with differenced values of d . Why would it be necessary to reach this degree of complexity? The ARIMA model helps predict future points in a time series and fits historical data well using this combination technique.

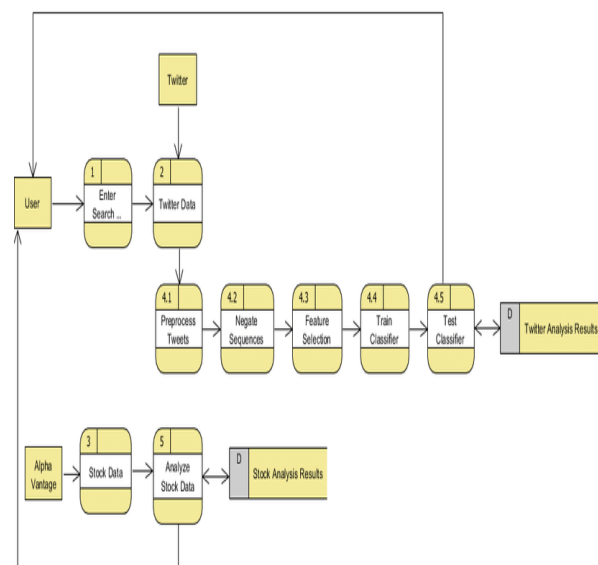
2. Random Forests: These ensemble learning models help in reducing overfitting and improving prediction accuracy by combining multiple decision trees (Patel et al., 2015). For investors and businesses listed on the stock market, the precise forecast or prediction of stock prices is a particularly important subject. Because stock prices are non-stationary and non-linear time-series, the prediction findings are extremely difficult and complex (Cavalcante et al., 2016). An extremely significant source of data for stock market forecasting is financial time series analysis (Oztekin et al., 2016). The prerequisite for analyzing and forecasting stock price movements is the discovery of hidden patterns. According to the well-known Random Walk hypothesis, the presumption According to the Efficient Market (Jensen M. C., 1978) and Malkiel, B. G. (2003), Mankiw & Shapiro, 1985), the existence of randomness and nonlinearity in the dataset makes it difficult to forecast the characteristics of the stock market. These hypotheses were confirmed by numerous pursuit models across varying time periods (Atsalakis & Valavanis, 2009).

3. Natural Language Processing (NLP): Sentiment analysis using NLP techniques on financial news, earnings reports, and social media data (e.g., Twitter, Reddit) has been integrated into stock prediction models (Bollen et al., 2011). By merging

computational linguistics—the rule-based modeling of human language—with statistical modeling, machine learning, and deep learning, natural language processing (NLP) allows computers and digital devices to detect, comprehend, and produce text and speech. From LLMs' communication abilities to picture generating models' comprehension of requests, NLP research has paved the way for the era of generative AI. Many people's daily lives now incorporate natural language processing (NLP), which powers search engines, voice-activated chatbots for customer support, voice-operated GPS systems, and smartphone digital assistants that can answer questions, like Microsoft's Cortana, Apple's Siri, and Amazon's Alexa.

4. Unsupervised Learning: Clustering techniques, such as K-Means and hierarchical clustering, have been used to group stocks with similar price movements for better prediction (Kou et al., 2021). K-Means is the clustering that is most frequently discussed in machine learning for novices. Finding c new observations—also known as centroids—randomly from our unlabeled data is the first stage. There are as many output classes as there are centroids. For every centroid, the iterative process begins with locating the closest point (based on Euclidean distance) and classifying it. The average of all the points assigned to each class is then calculated for each category. The new class centroid is the result.

5. CNNs: They have been adapted to analyze stock price data and financial news sentiment, improving predictive power (Chen et al., 2019). At the core of deep learning algorithms are neural networks, a subtype of machine learning. Their node levels consist of an input layer, an output layer, and one or more hidden layers. Every node has a threshold and weight that are connected to one another. Any node that produces an output above the designated threshold value is activated and sends data to the network's next tier. If not, no information is transmitted to the network's subsequent tier.



1. DATA FLOW DIAGRAM

III EXISTING SYSTEM

Traditional statistical models like GARCH and ARIMA have given way to more sophisticated AI-driven methods in stock price prediction systems. While machine learning techniques like Support Vector Machines (SVM), Random Forest, and Gradient Boosting increased prediction accuracy by making investment suggestions, early models still relied on time series forecasting and regression approaches. enabling data-driven decision-making. This system provides higher prediction accuracy, real-time market adaptation, and robust trading strategies, offering investors an intelligent tool to navigate the complexities of stock markets. analyzing complex market patterns. Deep learning techniques, By identifying sequential dependencies and obtaining characteristics from stock charts, algorithms such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) improved predictions even further. By examining investor sentiment, social media, and financial news, sentiment analysis employing Natural Language Processing (NLP) is also essential for forecasting market movements. Modern hybrid models combine statistical, machine learning, and sentiment analysis techniques to improve forecasting precision. Additionally, AI-powered trading systems, such as algorithmic and high-frequency trading, leverage real-time data and reinforcement learning for automated decision-making. These advancements have significantly improved the reliability of stock price predictions, but market volatility and external economic factors still pose challenges.

prediction model employs a hybrid deep learning approach, combining Long Short-Term Memory (LSTM) for temporal trend analysis, Convolutional Neural Networks (CNNs) for pattern recognition, and XGBoost for handling nonlinear relationships. Additionally, transformer-based models analyze financial news to assess market sentiment. To further optimize trading strategies, the system integrates a reinforcement learning-based Deep Q-Network (DQN) that dynamically adjusts buy and sell decisions to maximize returns while minimizing risk. A real-time dashboard displays stock forecasts, sentiment trends, and investment recommendations, enabling data-driven decision-making. This system provides higher prediction accuracy, real-time market adaptation, and robust trading strategies, offering investors an intelligent tool to navigate the complexities of stock markets.

V METHODOLOGY

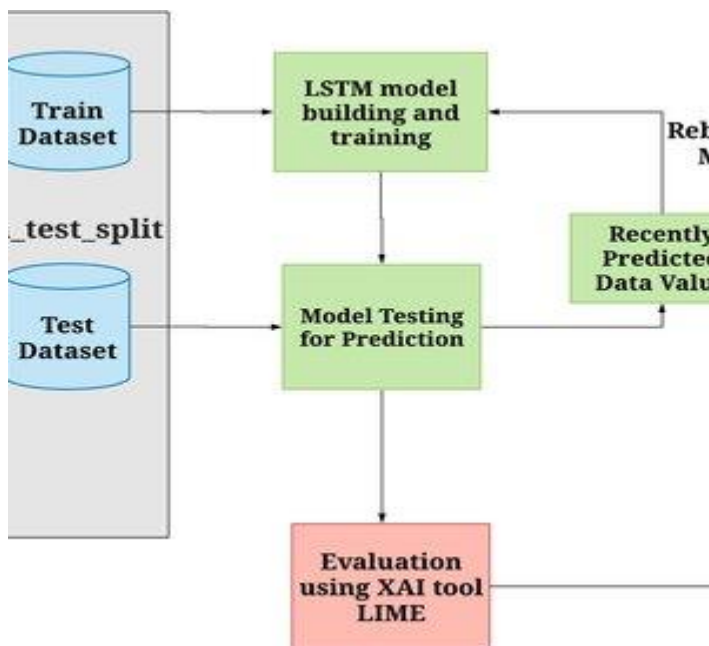
A. DATASET PREPARATION

Downloading, reorganizing, and structuring the stock price dataset for training and validation are all part of the dataset preparation step. The one that The dataset is authenticated and downloaded via the Kaggle API. The first phase is data gathering, which involves retrieving historical stock prices (Open, High, Low, Close, and Volume) from websites such as Quandl, Alpha Vantage, and Yahoo Finance. Additional datasets such as macroeconomic indicators, market sentiment, or news data can enhance predictions. Next, the data preprocessing step ensures data quality by handling missing values (using interpolation or forward-filling), removing duplicates, and normalizing numerical features to bring them within a standard range. The data is then split into training, validation and testing sets to avoid overfitting.

B. Data Preprocessing

Identified using statistical methods such as the IQR or Z-score, and then eliminated or modified. The next step is to apply data normalization. to scale numerical characteristics. Because stock prices might fluctuate greatly, methods such as Standardization (Z-score normalization) or Min-Max Scaling assist in bringing all features into a comparable range. *Standardization (Z-score normalization) or Min-Max Scaling assist in bringing all features into comparable range.*

improving model stability and convergence. Feature engineering is then performed by adding technical indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), Relative Strength Index (RSI), and Bollinger Bands. These features help models capture trends, momentum, and volatility patterns in stock prices. Additionally, time-based



IV PROPOSED SYSTEM

The proposed stock price prediction system enhances accuracy by integrating deep learning, machine learning, and sentiment analysis techniques. Unlike traditional models relying solely on historical prices, this system collects data from multiple sources, including financial news, social media sentiment, and macroeconomic indicators, to provide a comprehensive market outlook. It preprocesses data by removing noise, normalizing values, and extracting key features such as moving averages, volatility indicators, and sentiment scores derived from NLP models like BERT. The

features like day of the week, month, or trading session (morning/evening) can enhance predictive accuracy.

C. Model Training

The first step is choosing an appropriate model based on the prediction task. Traditional machine learning models like Linear Regression, Random Forest, and XGBoost work well for short-term trend predictions. However, deep learning models that are better at identifying temporal relationships in stock price movements include Transformers, Gated Recurrent Units (GRU), and Long Short-Term Memory (LSTM). After choosing a model, it is fitted using the training dataset that was created during preprocessing. The next day's closing price is usually the target variable in supervised learning, which uses engineering features and previous stock prices as inputs. Data is organized into sequences for LSTM and other time-series models using a sliding window technique. Optimization methods like Grid Search or Random Search (for machine learning) and Adam or RMSprop (for deep learning) aid in optimizing hyperparameters like learning rate, batch size, and number of layers during training in order to boost performance.

D. Model Evaluation

Model evaluation for stock price prediction is essential to assess accuracy, reliability, and generalization before deployment. Choosing suitable performance metrics is the first step in the evaluation process. These metrics include Mean Absolute Error (MAE) for a more comprehensible average error, Mean Squared Error (MSE) for assessing the magnitude of errors, R-squared (R^2) Score, which gauges how well the model explains price fluctuations, and Root Mean Squared Error (RMSE), which penalizes large errors. Reduce the RMSE and A high R^2 and MAE both point to a successful model. After that, an unseen validation/test dataset is used to verify the model's ability to generalize to new data.

In time-series models like LSTM, walk-forward validation is commonly used, where predictions are iteratively tested on sequential time windows to mimic real-world forecasting.

Additionally, techniques like cross-validation for traditional models and dropout regularization for deep learning help prevent overfitting. Finally, the model's robustness

is assessed under different market conditions, including volatile and stable periods, ensuring it performs reliably before being deployed for real-time stock price predictions.

VI CONCLUSION

In conclusion, while stock price prediction models demonstrate promising accuracy and reliability in controlled conditions, their real-world effectiveness remains subject to various challenges. The performance analysis indicates that models such as LSTM, ARIMA, and XGBoost can achieve reasonable error margins and explanatory power, with regression models showing RMSE values within acceptable limits and classification models

attaining moderate accuracy levels. The unpredictability of financial markets, which is impacted by outside variables including investor sentiment, geopolitical developments, and economic news, frequently restricts the consistency of forecasts, notwithstanding these encouraging outcomes. In order to enhance real-world applicability, backtesting and risk-adjusted performance indicators emphasize how crucial model validation and optimization are. Ultimately, while predictive models provide valuable insights for traders and investors, they should be used in conjunction with fundamental analysis, risk management strategies, and continuous monitoring to adapt to changing market *dynamics*.

VII REFERENCES

1. Research on Stock Price Prediction Method Based On Convolutional Neural Network, IEEE 2019- Sayavong Lounnapha.
2. Enhancing Profit by Predicting Stock Prices using Deep Neural Networks, IEEE 2019- Soheila Abrisham
3. An LSTM-Method for Bit-coin Price Prediction: A Case Study Yahoo Finance Stock Market, IEEE 2019- Ferdiansyah
4. Share Price Prediction using Machine Learning Technique, IEEE 2019- Jeevan B
5. Stock Market Prediction Using Machine Learning Techniques, IEEE 2020- Naadun Sirimevan
6. Stock Market Prediction Using Machine Learning 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC).

